MIL-B-37016A
8 August 1978
SUPERSEDING
MIL-B-37016 (DSA-DM)
24 October 1973

MILITARY SPECIFICATION

BOX, PLASTIC, INSULATED, Meat and Dairy Products, Laboratory Sample

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an insulated box, completed with a non-attached cover, for use by Veterinary Corps food inspection personnel, to provide sufficient insulation in the shipment of samples of perishable food products in a frozen state, from the point of collection to military laboratories, for analysis.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

H-P-621	- Brush, Stencil.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, CODE: ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8115

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:47Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D-2125 - Cellular Polystyrene, Specification for

(Application for copies should be sent to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

3. REQUIREMENTS

* 3.1 Material. The box shall be fabricated from polystyrene foam having a density range of 1.75 to 2.24 pounds per cubic foot, conforming to type I, class 1, grade 2 of ASTM Designation 2125-62T from 1968 Book of ASTM Standards, Part 26. The integrity of structure of the foam blocks (i.e., fusion of the expanding polystyrene beads) shall be such as not to release more than 12 beads per 100 square inches (sq. in.) when tested as specified (see 4.4.2).

3.2 Construction. The box shall consist of a rectangular body and a detachable plugged cover. Both components shall be of one piece molded construction. The cover shall be molded to form a top rectangular section with a rabbeted solid rectangular plug section underneath. When the cover is secured to the body, the plug section shall form a snug fit with the inside walls of the body, and the outer periphery shall be within +0, -1/16 inch of the actual outside length and width of the body walls.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:47Z
Check the source to verify that this is the current version before use.

3.3 Dimensions. The box shall conform to the following dimensional requirements (inches).

a. Cover	Top section	Plug section
Length and width	See 3.2	See 3.2
Thickness	1-1/8 ± 1/8	1/2 to 5/8
Radius of edges	--	1/8 ± 1/16

b. Body	Outside	Inside compartment
Length	11 ± 1/4	7-7/8 ± 1/4
Width	9 ± 1/4	5-7/8 ± 1/4
Height	10 ± 1/8	8-1/4 ± 1/4

* 3.4 Instructions for use. Each box shall be provided with a sheet form of instructions for use, imprinted as follows:

NOTE: THE FOLLOWING TABLE SHOULD BE USED AS A GUIDE. VARIATIONS IN COLLECTION METHOD, TYPE OF PRODUCT, SHIPPING FACILITIES, ETC. MAY NECESSITATE SOME DEVIATION FROM THE TABLE.

Outside temperature	No. hours in transit	Pounds of sample	Pounds of refrigerant (dry ice)
Below 50°F.	72	2	5
	48	3	4
	24	4	3
50 - 80°F.	72	2	5
	48	3	4
	24	3	4
80 - 100°F.	72	1	6
	48	2	5
	24	3	4
Over 100°F.	72	(Not recommended for shipments requiring more than 48 hours)	
	48	1	6
	24	2	5

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:47Z
Check the source to verify that this is the current version before use.

PRECAUTIONS

(a) Store box with cover off.
(b) Liquid samples should be placed in moisture-tight containers.
(c) Pre-chill open box for 24 hours prior to loading.
(d) Top of insulated box should be closed tightly with pressure sensitive tape when loaded.

3.5 Performance. Temperature inside box shall be not greater than -40°F., when tested as specified (see 4.4.1).

3.6 Finish. All surfaces (inside and outside) of the box and cover shall have a mold skin surface finish.

3.7 Identification marking. The name or registered trademark of the manufacturer shall be molded or permanently stamped on the bottom of each box.

3.8 Workmanship. Workmanship shall be first class throughout. Boxes shall be free from voids, cracks, chipped areas, odor and other defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:47Z
Check the source to verify that this is the current version before use.

4.1.3 Certificate of quality. Certificate of quality, supplied by the manufacturer of the polystyrene material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

* 4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I. The unit of product for sampling purposes shall be one insulated box.

TABLE I. Sampling for examination.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	2.5
Minor defects	II	4.0
For dimensional examination	S-2	2.5

* 4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II. The unit of product for sampling purposes shall be one insulated box.

TABLE II. Sampling for tests.

Characteristic	Requirement	Test procedure	Inspection level	AQL
Insulation	3.5	4.4.1	S-2	*
Integrity of structure	3.1	4.4.1	**	*

* Acceptance number shall be zero.
** One unit per lot.

4.3 Examination. The box shall be examined to determine compliance with all requirements contained in this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:47Z
Check the source to verify that this is the current version before use.

* 4.3.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects listed in table III. Examination shall not be restricted to the classified possible defects listed below.

TABLE III. Classification of defects.

Categories	Defects
Major	
101	Molded components not complete or not properly formed.
102	Box not free of fractures, splits, punctures, cuts or crushed areas.
103	Box not free of dirt or other contamination.
104	Box not free of accumulation of unexpanded material.
105	Box not free of warping or distortion.
106	Finish not continuous.
107	Cover flange does not properly set or is not flush on body of box.
108	Molded components not free of blisters.
109	Molded components not free from visible voids more than 1/16 inch across the largest dimension.
110	Molded components not free of moisture.
111	Cover does not fit snugly inside of box.
Minor	
201	Identification marking not correct, complete or legible.
202	Molded components not free of projections, burrs or sharp edges.

4.3.2 Dimensional examination. The boxes shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

* 4.4 Packaging inspection. The inspection of the unit, package, packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.5 Tests. Tests shall be conducted to determine compliance with specification requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:47Z
Check the source to verify that this is the current version before use.

* 4.5.1 Insulation. The open box shall be held at a temperature of -40°F. for 24 hours before testing. Four pounds (+0, -0.1 pound) dry ice (solidified carbon dioxide) and the sensing portion of a temperature-indicating device shall be placed in the box and the cover closed and taped as indicated in 3.4. The box shall be store at room temperature at 70°F. ± 2° for 24 hours. At the end of 24 hours, box shall be opened and temperature shall be read. Temperature may be indicated by a remote thermometer, a total immersion thermometer or by material which changes physical state at -40°F. Sensing portion of indicating device shall not be in contact with the dry ice. If remote thermometer is used, temperature may be read before opening the box.

* 4.5.2 Integrity of structure. A cover and samples of both 7-3/4 x 8-1/8 sections of the body shall be selected. Each specimen shall be broken into approximately two equal parts over a clean surface; and the two parts in turn shall be broken in a similar manner. Prior to breaking a specimen, one surface shall be scored to the minimum depth necessary to allow breaking by hand pressure. Any discrete beads loosened by fracturing shall be retained and segregated for each sample. The total fracture areas for each specimen shall be calculated in square inches. The fractured surfaces shall be subjected to a reciprocating stroke under normal painting pressure by use of a brush conforming to H-B-621. All released discrete beads shall be segregated and accumulated for each specimen, and shall be counted and calculated to determine the number of beads per 100 sq. in. surface area.

* 5. PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Unit of issue. One box along with instructions for use constitutes a unit of issue.

5.1.2 Level A. Each box shall be clean and dry with the lid in place. The box along with instructions for usee (see 3.4) shall be packaged in a box conforming to PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure shall be specified in the appendix of the box specification. As an alternate, each box may be wrapped in kraft paper and the wrap secured with kraft gummed tape.

5.1.3 Level C. Each box shall be clean, dry and packaged along with the instructions for use in a manner that will afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:47Z
Check the source to verify that this is the current version before use.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Four unit packages shall be packaged in a shipping container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Four unit packages shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged boxes shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the carriers' rules and regulations.

5.3 Marking. Each unit package and shipping container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. The subject item is intended for use inside of NSN 8115-00-183-9490, Box, Fiberboard, Corrugated.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) National Stock Number (NSN).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification covers the following item listed in the Federal Supply Catalog.

National Stock Number	Item Identification
8115-00-682-6525	BOX, PLASTIC, INSULATED, Meat and Dairy Products, Laboratory Sample

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:47Z
Check the source to verify that this is the current version before use.

6.4 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types normally procured to meet military requirements.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Logistics Agency - DM

Project Number 8115-0425

Review activity:
DoD - MB

122/4028

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:47Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:47Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-B-37016A BOX, PLASTIC, INSULATED, Meat and Dairy Products,

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER Laboratory Sample

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 202610-21T09:47Z
Check the source to verify that this is the current version before use.

FOLD

DEFENSE PERSONNEL SUPPORT CENTER
2800 SOUTH 20TH STREET
PHILADELPHIA, PA. 19101

POSTAGE AND FEES PAID

U.S.MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE MEDICAL MATERIEL, CODE: ATT
2800 SOUTH 20TH STREET
PHILADELPHIA, PA. 19101

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:47Z
Check the source to verify that this is the current version before use.